Linear Algebra in a Day (Or So)

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1 Introduction

1.1 What is Linear Algebra and Why Should We Care?

- Linear algebra is the branch of mathematics concerning the study of vector spaces, linear equations, and the mappings between those spaces.
- It is the mathematical framework that allows us to represent and solve systems of simultaneous equations, model physical phenomena, and interface with data.
- Within mathematics itself, it is a necessity to solve ODEs and to study differential geometry, among other use cases.
- In all of physics, the study of vectors and vector field transformations is embedded from the start and remains a much-needed tool for its duration.
- It was proven through the collective work of many physicists (and mathematicians) that linear algebra is the language of quantum mechanics.
  - Werner Heisenberg: Independently formulated the first consistent quantum framework in 1925 (matrix mechanics) using infinite arrays of numbers to describe atomic transitions
  - Max Born & Pascual Jordan: Recognized Heisenberg's numerical arrays as the matrices used in linear algebra, officially formalizing this into "matrix mechanics".
  - Paul Dirac: Unified the competing theories of matrix and wave mechanics into a single, elegant framework, inventing bra-ket notation.
  - John von Neumann: Provided the first complete, rigorous mathematical foundation of quantum mechanics in 1932, firmly establishing that quantum states live in Hilbert spaces.
  - Thus, our mission is clear: to understand quantum computing, we must understand quantum mechanics. As so, we must study Linear Algebra!

1.2 Where are We?

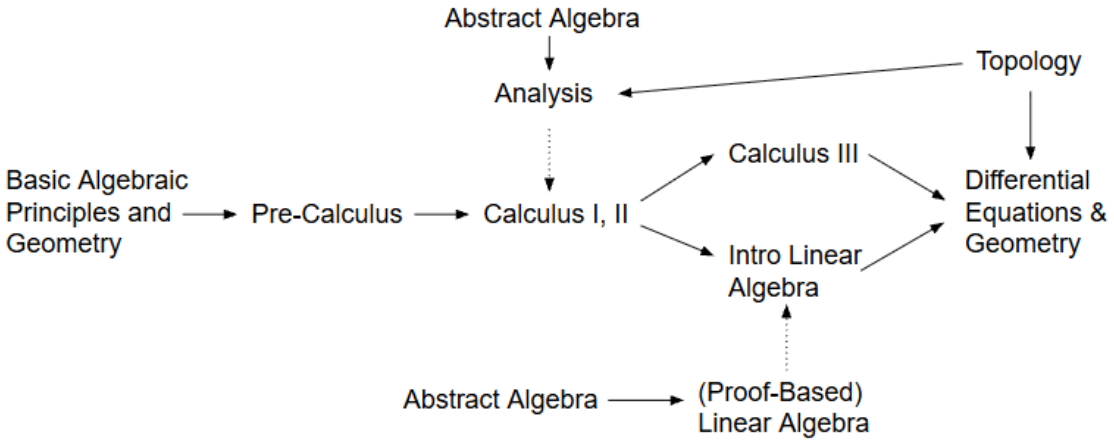


Figure 1: A Sparse Roadmap to Mathematics

- For perspective, above is a rough timeline of high school (and college) mathematics.
- I used a dotted line to convey a jump between proof-based mathematics and "practical" math.
- For this class, we will stick to the material of introductory linear algebra to skate by with

2 Fields in Abstract Algebra

- As indicated on our roadmap, to start linear algebra, we must have an understanding of (abstract) algebra.
- I will assume we are familiar with the elementary algebra of real and complex numbers for the sake of time.
- Luckily, we don't need much out of abstract algebra, just the definition of a field (and homomorphisms) to construct our concept of vectors.
- **Fields**: a **set** equipped with two binary operations that satisfy specific rules called field axioms. These axioms are required to hold for all elements of our field:
  1. Closure for finite sums and products
  2. Associativity of addition and multiplication
  3. Commutativity of addition and multiplication
  4. Additive and multiplicative identities (1 and 0)
  5. Additive and Multiplicative inverses (note that the 0 element is an exception to the rule of multiplicative inverses)
  6. Distributivity of multiplication over addition
- In the real (and complex) number field, we say $(+, \cdot, \mathbb{R})$ or $(+, \cdot, \mathbb{C})$. We can trivially show that our real and complex numbers satisfy these rules.
- Another example, that is simple, but distinct from the real numbers, is the rational numbers.
- **Characteristic of a Field**: it is possible to define the product $n \cdot a$ of an arbitrary element a of F by a positive integer n to be the n-fold sum $(a + a + ... + a$, which is an element of F). If no positive integer exists such that $n \cdot 1 = 0$, then F has characteristic 0; else, it has characteristic n. (n will always be prime)

  - This separates our idea of fields into two groups: finite and infinite fields. We are already familiar with infinite fields in our examples above.
  - We can construct basic finite fields by taking the integers and taking the modulus over a prime number. The most basic example is the prime 2. This is known as the binary field, and we can prove it for general p, but we won't for the sake of time.
  - **Order of a Field**: The number of unique elements in the field.
  - The astute of you may ask: is this not the same as characteristic? The more astute of you will answer that there exist finite field extensions.

- **Field Extensions**: Let F and K be fields. We say that K is an extension field of F, denoted by K/F, if F is a subfield of K.
  - We can very easily see that $\mathbb{C}$ is an extension of $\mathbb{R}$ since if we disregard the imaginary part of a complex number, we get back real arithmetic.
  - **Degree of a field extension**: the dimension of F as an E-vector space. It satisfies the formula: $[G : E] = [G : F][F : E]$. Think about this as "how many copies of the base field are present in the extension".
  - Our previous example is a field extension of degree two. (We have two copies of $\mathbb{R}$, one for the real part, one for the imaginary part)
  - To answer the difference between the characteristic and the order of a field. For a prime finite field, we can construct a finite extension such that the characteristic will remain prime, but the order will be a power of the prime (e.g. $F_2[x]/(x+1) \cong F_4$)

- **Field Homomorphisms**: Let F and K be fields. A *field homomorphism* is a map $\varphi : F \to K$ such that for all $a, b \in F$, $\varphi(a+b) = \varphi(a) + \varphi(b), \varphi(ab) = \varphi(a)\varphi(b)$, and $\varphi(1_F) = 1_K$.
  - This gives us a way to compare operation structures between fields. That is, if we have a field homomorphism, we have a way to relate the operations of one structure to those of another
  - **Injectivity and Surjectivity**: A function $f : A \to B$ is *injective* if $f(a_1) = f(a_2)$ implies $a_1 = a_2$, and *surjective* if for every $b \in B$ there exists $a \in A$ such that $f(a) = b$.
  - We can easily argue that all field homomorphisms are injective. Suppose $f(a) = f(b)$. then $f(a) - f(b) = f(a-b) = 0$. Assume for the sake of contradiction that $a - b \neq 0$, then $f(a-b)f((a-b)^{-1}) = 1 = 0 \cdot f((a-b)^{-1})$, which is not true since there are no zero divisors in a field. Thus, $a = b$.
  - If we can show a function is both injective and surjective, it is bijective. If our bijective function is a field homomorphism, we get a field isomorphism. What this tells us is that arithmetic in one field is directly analogous to arithmetic in another.
  - All fields of the same order are isomorphic. Since we can easily construct all field extensions of all orders, we know very well how any given field will behave just by its order.

3 Matrix Math

- In your Algebra II classes, you may have seen your teacher represent a system of equations with an $n \times m$ matrix, then, through magic, solve the system using RREF. We will learn why precisely this works later, but let's do a quick review of the math behind it.

3.1 Matrix Algebra

- We have well-defined rules for matrix addition, subtraction, and multiplication (we will discuss "division" later).

- **Matrix Addition:** We can add two $n \times m$ matrices together with component-wise: For $n \times m$ matrices $A, B \in M_F^{n \times m}$, $(A+B)_{ij} = A_{ij} + B_{ij}$. Subtraction is simply the opposite of addition. Here, the zero matrix acts like the additive identity $(0_{ij} = 0, \forall i, j)$

- **Matrix Multiplication**: Matrix multiplication AB is well-defined when A is an $m \times n$ matrix and B is an $n \times p$ matrix (where n,m,p are natural numbers). We say $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$. This formula says that the (ij)-th position is the dot product between the i-th row and the j-th column. In matrix multiplication, the identity matrix acts as the multiplicative identity ($I = \delta_{ij}$).
- Note: while matrix addition is commutative, matrix multiplication is not. Also, cancellation generally fails (unless an inverse exists)
- As a result of the failure of commutation, $A = B \implies CA = CB$, but not $CA = BC$.
- **Inverses (Matrix Division?):** Certain matrices have inverses; we will identify later. That is $\exists A^{-1}$, such that $AA^{-1} = A^{-1}A = I$. We can use inverses, if they exist, to "cancel" matrices. But before we do, we must learn a few rules.
  1. To "cancel" matrices with their inverse, they must be adjacent to their inverse ($A^{-1}BA \neq B$)
  2. The inverse is unique, so $(A^{-1})^{-1} = A$. And the inverse of a scalar is its reciprocal.
  3. $(AB)^{-1} = B^{-1}A^{-1}$
- **Transposes:** The transpose flips an $n \times m$ into $m \times n$ matrix, reflecting each entry along its diagonal. **Symmetric Matrices** are invariant under transposes. Properties include:
  1. $(A^T)^T = A$
  2. $(AB)^T = B^T A^T$
  3. $(A^T)^{-1} = (A^{-1})^T$
- **Exponents of Matrices:** A^n is simply n-successive multiplication. Note: $(AB)^2 \neq A^2B^2, (AB)^2 = ABAB$. While there exist exponent expressions where we have e^A, this is shorthand for the Taylor expansion.
- **Trace:** The sum of entries on the diagonal of a matrix. The trace of a matrix is equal to the sum of its eigenvalues (learn more later), and is invariant under similarity transformations. We have the following trace properties:
  1. $tr(A + B) = tr(A) + tr(B)$
  2. $tr(cA) = c \cdot tr(A)$
  3. $tr(A^T) = tr(A)$
  4. Traces are equal under cyclic permutations of the product of matrices

3.2 Reduced Row Echelon Form

- **Elementary Row Operations:** three basic operations that can be performed on the rows of a matrix without changing the solution set (or elementary canonical form) of the corresponding system of linear equations:
  1. Row interchange: $R_i \longleftrightarrow R_j$
  2. Row scaling: $R_i \to cR_i, c \neq 0$
  3. Row replacement: $R_i \to R_i + cR_j, i \neq j$
- **Elementary Column Operations:** replace the word row with column in the above definition
- **Reduced Row Echelon Form:** A canonical matrix (equivalent by elementary row and column operations) that is obtained by performing elementary row operations until the matrix satisfies:
  1. All zero rows are at the bottom.
  2. The first nonzero entry (the pivot) of each nonzero row is 1
  3. Each pivot lies strictly to the right of the pivot in the row above it.
  4. Each pivot is the only nonzero entry in its column.

- We can use this to solve linear systems of equations because elementary row operations don't alter the solution space, so once in RREF, solutions will be easily realized.
- The rank (see definition in a section below) is the number of pivot columns
- Columns without pivots are linear combinations (see below) of the pivot columns
- An $n \times n$ matrix is invertible iff its RREF is the identity matrix.

4 Describing Vector Spaces

- We have finally escaped abstract algebra and are ready to tackle its linear friend.
- You may be used to the notion of a vector from your calculus or physics high school class, an object with both direction and magnitude, right? Well, not really. We will learn in this section what vectors are and where they live.

4.1 Vector Spaces

- **Vector Spaces**: A pair of sets (V, F), where V is a set of vectors, and F is a (scalar) field. This pair must satisfy two properties: vector addition and scalar multiplication of vectors:
  1. **Vector Addition**: $\vec{v_1}, \vec{v_2} \in V \implies \vec{v_1} + \vec{v_2} \in V$
  2. **Scalar Multipication**: $c \in F, \vec{v} \in V \implies c\vec{v} \in V$
  3. Closure under finite addition and multiplication
  4. Addition is commutative and associative
  5. There exists a unique vector called the zero vector that, when added to another vector, yields the original vector
  6. For each vector, there is an opposite vector, such that when the two are added together, they yield zero.
  7. The multiplicative identity of the field will yield the same vector when scalar multiplied by a given vector
  8. Scalar multiplication is associative and distributive over vector addition
  9. Scalar multiplication is distributive over scalar addition ($(c_1 + c_2)\alpha = c_1\alpha + c_2\alpha$)
- Often, when there is no confusion as to the vector field for which our vectors may belong, we simply say that our V is a vector space over the field F.
- The definition of a vector space is a bit vague, and this is intentional to capture the fullest extent of what vectors can do. This can allow us to create funny vector spaces, such as the vector space of groceries over the field of rational numbers, allowing you to write down your entire grocery list with a single well-placed vector.
- Our notion of what defines a vector is now incredibly broadened to be any object that lies inside a vector space.
- Let us now discuss some examples of vector spaces that we may come across:
  - The n-Tuple Space, F^n, over Field F: For some field F, we define the vectors of this vector space as all tuples of the form: $\alpha = (x_1, x_2, ..., x_n)$, for $x_i \in F$. Vector addition is defined component by component, and scalar multiplication distributes over all entries.
  - The Set of All Polynomials of Degree n or Lower, $F_n[x]$: The vectors here, as you can guess, are all the polynomials of degree n or less. Vector addition is polynomial addition, and scalar multiplication is simply multiplication by an element of the field (not a polynomial).
- **Linear Combinations:** A vector, β, in V is said to be a linear combination of the vectors $\alpha_1, \alpha_2, ...$ in V if there exist scalars $c_1, c_2, ...$ such that: $\beta = \sum^n c_i\alpha_i$

- **Subspaces**: As the name seems to suggest, *subspace* is a vector space inside another vector space over the field F with vector addition and scalar multiplication inherited from the vector space it resides in.
- An example of a subspace is a plane embedded in 3D space. We say that the plane (regardless of how it is oriented) forms a 2-dimensional subspace in $\mathbb{R}^3$. Note, you lose a lot of information when you project into a subspace. Here, we lose the concept of width in our subspace (all vectors now lie in a flat plane).
- **Span**: Let S be a set of vectors in a vector space V. The subspace spanned by S is the intersection W of all subspaces of V that contain S. This definition simplifies nicely to be the vector space W formed by the set of all linear combinations formed by vectors in S.
- We can consider the sum of two subspaces W_1, W_2 as $W_{1+2} = W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$. This is different than just considering the union of the two sets (Think about the plane in $\mathbb{R}^3$ example; the union of two intersecting but not identical planes is simply the two planes, but the sum spans all of $\mathbb{R}^3$)

4.2 Bases, Dimension, and Coordinates

- **Linear Independence**: A set of vectors, $\beta = \{\beta_1, \beta_2...\}$ is considered to be linearly independent iff $\sum^n c_i\beta_i = 0 \implies c_i = 0, \forall\ i \leq n$. We say the set is linearly dependent if not linearly independent.
- Any subset of a linearly independent set is linearly independent
- **Basis:** Let V be a vector space. We say a basis for V is a linearly independent set of vectors that span the space V. We say the space V is finite-dimensional if the basis has a finite number of vectors.
- **Canonical**: You might hear the word canonical thrown around; it simply means "basic" or "independent of choice." Sometimes it also means without the consideration of bases. This is needed since certain infinite-dimensional spaces cannot have an explicit construction of a basis.
- **Dimension of a vector space V**: If V is finite-dimensional, the dimension of V is simply the number of vectors in the basis.
- Something to note: you can have multiple bases to represent a single vector space. A problem in linear algebra/physics may become significantly easier if you can choose the correct/suitable basis.
- A common basis we use to represent $\mathbb{R}^n$ is called the standard basis: e_i, where e_i is an n-tuple that is zero in all entries other than the i-th entry, which contains a 1. For $\mathbb{R}^3$, we use $\hat{i}, \hat{j}, \hat{k}$
- For a subspace W of V, we can choose a basis for the subspace, then extend the subspace to higher dimensions by adding linearly independent vectors (until we get to V itself).
- Subsets of bases span subspaces, but for a fixed basis, subspaces aren't always spanned by subsets of that fixed basis.
- One of the useful features of a basis in an n-dimensional vector space V is that it essentially enables one to introduce coordinates in V analogous to natural coordinates x_i of a vector $\alpha = (x_1, ..., x_n)$ in the space F^n
- **Ordered Basis**: If V is a finite-dimensional vector space, then an ordered basis for V is a finite sequence of vectors that is linearly independent and spans V.
- This is just a way to say that we fix the order that we discuss basis vectors and swapping the positions of two basis vectors yields a new ordered basis (even if the vectors themselves are identical), allowing us to talk about coordinates without being confused about which coordinate relates to which vector.

- **Coordinates Relative to an Ordered Basis:** Suppose V is a finite-dimensional vector space over the field F and that $\beta = \{\alpha_1, ..., \alpha_n\}$ is an ordered basis for V. Given an arbitrary vector $\alpha \in V$, there is a unique n-tuple of scalars in F such that $\alpha = \sum^n x_i\alpha_i$. Sometimes we write this as an upright column vector $[\alpha]_\beta = [x_1, ..., x_n]^T$

- Let V be an n-dimensional vector space over the field F, and let β and β' be two ordered bases of V. Then there is a unique, invertible, n X n matrix P with entries in F such that for any α:

  1. $[\alpha]_\beta = P[\alpha]_{\beta'}$
  2. $P^{-1}[\alpha]_\beta = [\alpha]_{\beta'}$

  The columns of P are given by $P_j = [\alpha'_j]_\beta,\ j = 1, ..., n$, where $[\alpha'_j]_\beta$ means the j-th basis vector of β' written as a linear combination of the vector in β with the column being the coefficients.

4.3 The Direct Sum

- The direct sum is a way of combining vector spaces while keeping their directions independent. Intuitively, it "adds" new dimensions.

- **Direct Sum**: Given U and W of a vector space V, we say that $V = U \oplus W$ if each vector $v \in V$ can be written uniquely as $v = u + w, u \in U,\ w \in W$.

- The word "uniquely" is crucial. A sum of subspaces is a direct sum precisely when the subspaces are disjoint (other than the zero vector). That is, they share no non-zero directions. An equivalent condition is that $u + w = u' + w'$ then necessarily $u = u'$ and $w = w'$.

- If $V = U \oplus W$ and V is finite dimensional then $dim(V) = dim(U) + dim(W)$. Thus, direct sums add dimensions.

- If $\{u_1, ..., u_m\}$ is a basis of U and $\{w_1, ..., w_n\}$ is a basis for W their union forms a basis for V.

- The notion of a direct sum extends naturally to infinitely many vector spaces.

- **Infinite Direct Sum**: *The infinite direct sum* of a collection of vector spaces $\{V_i\}_{i\in I}$, denoted by $\bigoplus_{i\in I} V_i$, consists of all tuples $(v_i)_{i\in I}$ such that $v_i \in V_i$ and only finitely many of the v_i are nonzero. The requirement of finite support ensures that each vector has a finite, unique decomposition into components from the individual spaces.

- There is a version with infinite support called the direct product. They are equivalent in finite dimensions, but their properties diverge in the infinite dimensional case.

5 Linear Transformations and Dual Spaces

- Now that we have a good understanding of what a vector space is, let us discuss transformations between vector spaces.

5.1 Linear Transformations:

- Let V and W be vector spaces over the field F. A linear transformation from V to W is a function, *f*, from V into W such that: $T(c\alpha + \beta) = c(T(\alpha)) + T(\beta)$ for all $\alpha, \beta \in V$ and all scalars c in F.

- The most basic example that I can think about is the zero transformation, where T sends every vector from V to the zero vector in W.

- Suppose V has a basis β. We say that every linear transformation, $T : V \to W$, is uniquely defined by its action on the basis vectors of V. Since, by the linearity of our function, we can break the action of our linear transformation to act on our individual basis vectors: $T(\alpha) = T(\sum c_i\beta_i) = \sum c_iT(\beta_i)$.

- As a result of the above statement, for vector spaces V with ordered basis $\beta = \{\beta_1, ..., \beta_n\}$ and W with vectors $\alpha_1, ..., \alpha_n$, we see that there is precisely one linear transformation such that $T\alpha_j = \beta_j$ for $1 \leq\ j \leq n$
- **Null Space (Kernel):** Let V and W be vector spaces over the field F, and let T be a linear transformation from V into W. The *null space* of T is the set of all vectors such that $\mathrm{T}(\alpha) = 0$
- **Rank and Nullity:** If V is finite-dimensional, the *rank* of T is the dimension of the range of T, and the *nullity* is the dimension of the null space of T.
- **The Rank Nullity Theorem:** Let V and W be vector spaces over the field F and let T be a linear transform from V into W. Suppose that V is finite-dimensional. Then, $rank(T) + nullity(T) = dim(V)$.
- The set of linear transformations from vector spaces V to W forms a vector space. We define vector addition as the addition of two functions ($(T + U)(\alpha) = T(\alpha) + U(\alpha)$), and scalar multiplication is self-explanatory.
- The space of all linear transformations between finite-dimensional V (n) and W (m), L(V, W), is a finite, $n \cdot m$-dimensional vector space. This makes sense as, if we have bases β_V and β_W, there exists nm possible ways to match each input to an output (where one or more basis vectors can be mapped to a single basis vector). The sum and the scaling of these linear basis functions can then be used to reproduce any target linear function.
- We can also compose linear functions. Take $S : U \to V$ and $T : V \to W$, for vector spaces U, V, W, we can define $TS : U \to W$ as $(TS)(\alpha) = T(S(\alpha)), \forall \alpha \in R$
- **Linear Operator (Endomorphism):** We say that a linear transform $T : V \to V$ is called a *linear operator*. An example could be a change-of-bases mapping. The set of all endomorphisms forms a ring (a field without support for division, e.g. the set of integers).
- **Invertible Transforms:** A linear function is invertible iff it is both surjective (onto) and injective (one-to-one). We also call this bijective linear transform an isomorphism (if such exists between two vector spaces, we call the pair isomorphic).
- For an isomorphism $T : V \to W$, we indicate the inverse transform of this function as $T^{-1} : W \to V$. If this is an isomorphism between two distinct vector spaces, this tells us (like for a field isomorphism) that the underlying algebraic structure of the vector spaces is identical.
- If we have a linear transform $T : V \to W$ with $dim(V) = dim(W)$ (for finite dimensions), the following statements are equivalent (by reasoning of the rank-nullity theorem):

  1. T is invertible
  2. T is injective
  3. T is surjective

- Every n-dimensional vector space over the field F is isomorphic to the space F^n by choosing coordinates. Taking this one step further, all n-dimensional vector spaces are also isomorphic (but not all linear transforms between them are isomorphisms)!
- **Inclusion and Projection Maps:** For vector space V and subspace W. There exists a natural injective inclusion ($i : W \to V$) and surjective projection ($\pi : V \to W$) map. The inclusion maps each vector in W to itself in V. We define the projection map linearly: for a basis of W and an extended basis of V, we map each vector in the extended basis of V that is a basis element in W to itself and all other basis vectors to the zero vector in W.
- A natural consequence is that there exist natural projection and inclusion operators for direct sums (inclusion is self-explanatory). For $V = U \oplus W$, the projection of V onto U is $\pi_U : V \to U$ ($\pi_W : V \to W$), where $\pi_U(u + w) = u$ and $\pi_W(u + w) = w$. We also have that (if we redefine the function to be an endomorphism) $\pi_U + \pi_W = I$

5.2 Representations of Transformations by Matrices

- We discussed earlier the properties of matrices and what row reduction is, but now we learn what it all means
- Given V (n-dimensional) and W (m-dimensional) over field F, with ordered bases $\beta = \{\alpha_1, ..., \alpha_n\}$ for V and $\beta' = \{\beta_1, ..., \beta_m\}$ for W, any linear transformation T, as we have identified before, is fully determined by where it sends basis vectors: $T\alpha_j = \sum_{j=1}^{m} A_{ij}\beta_i$.
- The scalars A_{ij} form an $m \times n$ matrix A, relative to the given bases. Where its j-th column is the coordinate vector $[T\alpha_j]_{\beta'}$
- If $X = [\alpha]_\beta$, then $AX = [T\alpha]_{\beta'}$. Therefore, for a fixed pair of bases, given a linear transformation, there exists a unique $m \times n$ matrix over F, and vice versa: $L(V, W) \cong \{all\ m \times n\ matrices\}$
- For endomorphisms, we often just use a single basis (unless intentionally changing coordinates). Thus, for $T : V \to V$, $T\alpha_j = \sum_{ij} A_{ij}\alpha_i$ yields $[T]_\beta$ and satisfies $[T\alpha]_\beta = [T]_\beta[\alpha]_\beta$
- We can model the composition of linear functions as the product of matrices.
- **Similar Matrices/Transformations** Let A and B be n X n (square) matrices over the field F. We say that B is similar to A over F if there is an invertible n X n matrix P over F such that $B = P^{-1}AP$.

5.3 Linear Functionals

- **Linear Functionals**: If V is a vector space over the field F, a linear transformation f from V into the scalar field F is also called a linear functional on V.
- The set of these functions is trivially a vector space, called the dual space, $V* = L(V, F)$. This is intuitive, since the field itself is isomorphic to a one-dimensional vector space, and there is a corresponding linear transformation between V and this one-dimensional space.
- If V is finite-dimensional, we see that dim(V*)=dim(V) for a similar argument above.
- For example, let F be a field and let $a_1, ..., a_n$ be scalars in F. Define a function f on F_n by $f(x_1, ..., x_n) = a_1x_1 + ... + a_nx_n$ or in the standard basis, $a_j = f(\epsilon_j), j = 1, ..., n$.
- Given a basis $\beta = \{\alpha_1, ..., \alpha_n\}$ for V, there exist a unique dual basis $\beta^* = \{f_1, ..., f_n\}$ for V^*, such that $f_i(\alpha_j) = \delta_{ij}$, allowing us to construct any functional ($f = \sum f(\alpha_i)f_i$) and any vector ($\alpha = \sum f_i(\alpha)\alpha_i$. These f_i are the coordinate functions relative to β because the Kronecker delta effectively "picks out" the coefficients from the vector.
- **Annihilator:** If V is a vector space over the field F and S is a subset of V, *the annihilator* of S is the set S° of linear functionals f on V such that $f(\alpha) = 0$ for every α in S.
- The annihilator of S forms a subspace of V^*. If S=V, then S° is the zero subspace of V^*.
- For a subspace $W \subset V$, $dim(W) + dim(W^\circ) = dim(V)$. Thus, if W_1 and @ are subspaces of a finite-dimensional vector space, then $W_1 = W_2$ if and only if $W_1^\circ = W_2^\circ$.
- A homogeneous system AX=0 is exactly the problem of finding the subspace annihilated by the functionals defined by A's rows. Row-reducing A gives a basis for the solution space; dually, given vectors spanning W, row-reducing the matrix of those vectors as rows gives a basis for the annihilator W°.
- Each $\alpha \in V$ induces a functional L_α on V^* via $L_\alpha(f) = f(\alpha)$.
- The map $\alpha \mapsto L_\alpha$ is an **isomorphism of** V **onto** V^{**} (for finite-dimensional V): $V^{**} \cong V$
- Every functional on V^* is "evaluation at some $\alpha \in V$." Therefore, every basis of V^* is the dual of some basis of V.
- $(S^0)^0$ equals the subspace spanned by S.

5.4 Transposes

- We often imagine the transpose as the reflection of a matrix about its diagonal, but let us take a moment to actually describe it.
- Let V and W be vector spaces over the field F. For each linear transform from V into W, there is a unique linear transformation T^t from W^* into V^* such that: $(T^t g)(\alpha) = g(T\alpha)$, for every g in W^* and α in V.
- Let V and W be vector spaces over the field F, and let T be a linear transformation from V into W. The null space of T^t is the annihilator of the range of T. If V and W are finite-dimensional, then: $rank(T^t) = rank(T)$ and the range of T^t is the annihilator of the null space of T.
- In matrix form, we say that for matrix A, $A^t_{ij} = A_{ji}$.
- Let A be any $m \times n$ matrix over the field F. Then the row rank of A is equal to the column rank of A.

6 Determinants

- We will now discuss the most confusing part for most mathematicians' first look at the subject: determinants. There might be a part of you asking: what is hard about determinants, just do $ad-bc$ or cofactor expand, and you're done. But I want to talk about what exactly a determinant is.

6.1 Baby's First Determinant

- **Geometry's Determinant:** A number built from the entries of a square matrix, telling you whether the matrix is invertible and by how much it scales the volume of an n-dimensional box, that is invariant under choice of basis.
- A two-by-two determinant is often given by the formula: $det(A) = ad - bc$ for matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$
- **The General (Leibniz) Formula:** $det(A) = \sum_{\sigma \in S_n} A_{1,\sigma(1)} A_{2,\sigma(2)} ... A_{n,\sigma(n)}$. Broken down: we sum over all permutations σ **of** $\{1, ..., n\}, sgn(\sigma) = \pm 1$. This sum has $n!$ terms (you will seldom use this definition).
- **Cofactor Expansion:** Define the **minor**, M_{ij}, as the determinant of the matrix left after deleting the i-th row and j-th column, and the **cofactor**, $C_{ij} = (-1)^{i+j} M_{ij}$. Our cofactor expansion (along row i and column j respectively) is $det(A) = \sum_{j=1} A_{ij} C_{ij} = \sum_{i=1} A_{ij} C_{ij}$. Note, we get an alternating pattern given by the exponent of -1.
- **Algebraic Properties of Determinants:**
  1. $det(I) = 1$
  2. $det(AB) = det(A)det(B)$
  3. $det(A^{-1}) = 1/det(A)$
  4. $det(A^T) = det(A)$
  5. $det(cA) = c^n det(A)$ for an $n \times n$ matrix
  6. $det(A) \neq 0 \iff A\ non-singular \iff rank(A) = n$
- An important consequence of the determinant product rule is that the determinant is invariant under all similar matrices (change of bases). Analogous to curvature in differential geometry.
- Determinants are invariant under row operations. However, determinants are negated under row swaps, and scale inversely when we scale a row.

6.2 An n-Linear Alternating Form Construction

- This definition leaves much to be desired and more questions than answers: where do these properties emerge from, why is the formula so odd, and what does this mean? We can answer this with an explicit construction of a determinant using alternating forms.
- Let V be an n-dimensional vector space over a field F (this presupposes nothing else). Consider the space of alternating **n-linear forms on V**: $D : V \times V \times ... \times V \to F$ (n copies). These are linear in each argument separately, and alternating: swapping any two values negates the value. Any identical argument yields 0; this is precisely why if we have a matrix whose $rank < n$, the determinant of the matrix is 0.
- Given $T : V \to V$ linear, and any non-zero alternating n-form D, define the alternating n-form D_T by: $D_T(v_1, ..., v_n) := D(Tv_1, ..., Tv_n)$. Since the space of such one forms is 1-dimensional, $D_T = c \cdot D$ for some $c \in F$. Precisely, $c = det(T)$.
- If we choose $v_i = e_i$, we can precisely obtain the Leibniz formula. This is precisely the tool that we take so often for granted.

7 Tensors

- **Bilinear Maps**: If V, W, and U are vector spaces over a field F, a map $\varphi : V \times W \to U$ is **bilinear** if it is linear in each argument separately, holding the other fixed:

$$\varphi(av_1 + v_2, w) = a\varphi(v_1, w) + \varphi(v_2, w), \qquad \varphi(v, aw_1 + w_2) = a\varphi(v, w_1) + \varphi(v, w_2).$$

  Note that φ is *not* linear on $V \times W$ as a whole (that space has dimension $\dim V + \dim W$, not $(\dim V)(\dim W)$); bilinearity is a strictly weaker condition than linearity on the product space.

- **The Tensor Product**: The tensor product $V \otimes W$ is the vector space constructed so that bilinear maps out of $V \times W$ correspond exactly to linear maps out of $V \otimes W$. There is a fixed bilinear map $\otimes : V \times W \to V \otimes W$, $(v, w) \mapsto v \otimes w$, with the property that for every bilinear map $\varphi : V \times W \to U$, there is a *unique* linear map $\tilde{\varphi} : V \otimes W \to U$ such that $\varphi(v, w) = \tilde{\varphi}(v \otimes w)$ for all v, w. This is called the **universal property** of the tensor product, and it is what makes $V \otimes W$ the "correct" space for linearizing bilinear data.

- **Basis and Dimension**: If $\mathfrak{B} = \{e_1, \ldots, e_n\}$ is a basis for V and $\mathfrak{B}' = \{f_1, \ldots, f_m\}$ is a basis for W, then

$$\{e_i \otimes f_j : 1 \le i \le n,\ 1 \le j \le m\}$$

  is a basis for $V \otimes W$. Consequently,

$$\dim(V \otimes W) = (\dim V)(\dim W) = nm.$$

  This is the central fact distinguishing the tensor product from the direct sum, where $\dim(V \oplus W) = \dim V + \dim W$. Dimension *multiplies* under $\otimes$, rather than *adding* as it does under $\oplus$.

- **Bilinearity Rules**: For $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$, and $a \in F$, the symbol $\otimes$ satisfies

$$\begin{aligned} (av) \otimes w &= a(v \otimes w) = v \otimes (aw), \\ (v_1 + v_2) \otimes w &= v_1 \otimes w + v_2 \otimes w, \\ v \otimes (w_1 + w_2) &= v \otimes w_1 + v \otimes w_2. \end{aligned}$$

  These are exactly the defining relations of the tensor product: $V \otimes W$ is, informally, the vector space spanned by symbols $v \otimes w$ subject only to these bilinearity identities and no others.

- **Simple versus General Tensors**: An element of the form $v \otimes w$ for some particular $v \in V, w \in W$ is called a **simple** (or **decomposable**) tensor. A general element of $V \otimes W$ is a finite sum

$$\tau = \sum_{k=1}^{r} v_k \otimes w_k,$$

and not every such τ can be rewritten as a single simple tensor $v \otimes w$. An element that cannot be so rewritten is called **entangled** (non-factorizable); this is the purely linear-algebraic origin of the notion of entanglement in quantum mechanics.

- **Tensor Products of Linear Maps**: Given linear maps $T : V \to V$ and $S : W \to W$, there is a unique linear map

$$T \otimes S : V \otimes W \to V \otimes W, \qquad (T \otimes S)(v \otimes w) = Tv \otimes Sw,$$

extended to all of $V \otimes W$ by linearity. Existence and uniqueness of $T \otimes S$ again follow from the universal property, since $(v, w) \mapsto Tv \otimes Sw$ is bilinear in (v, w).

- **The Kronecker Product**: For an $m \times n$ matrix A and a $p \times q$ matrix B, the **Kronecker product** $A \otimes B$ is the $mp \times nq$ block matrix formed by replacing each entry A_{ij} with the block $A_{ij}B$:

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ A_{21}B & A_{22}B & \cdots & A_{2n}B \\ \vdots & \vdots & & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mn}B \end{bmatrix}.$$

- **Trace and Determinant Identities**: For $T : V \to V$ and $S : W \to W$ with $\dim V = n$, $\dim W = m$,

$$\operatorname{tr}(T \otimes S) = \operatorname{tr}(T)\operatorname{tr}(S), \qquad \det(T \otimes S) = (\det T)^m (\det S)^n.$$

These follow from the block structure of the Kronecker product matrix and serve as useful sanity checks in computation.

- **Inner Product on $V \otimes W$**: If V and W are each equipped with an inner product, $V \otimes W$ inherits a natural inner product, defined on simple tensors by

$$\langle v_1 \otimes w_1,\ v_2 \otimes w_2 \rangle = \langle v_1, v_2 \rangle \langle w_1, w_2 \rangle,$$

and extended to general elements of $V \otimes W$ by linearity in one slot and conjugate-linearity in the other. This is what allows norms, orthonormal bases, and (in the quantum context) probabilities to be defined on the composite space $V \otimes W$.

- **Connection to Bilinear Forms**: Just as the dual space V^* organizes linear functionals on V, the space $(V \otimes W)^*$ organizes bilinear functionals on $V \times W$; in fact $(V \otimes W)^* \cong V^* \otimes W^*$ for finite-dimensional V, W, paralleling the way $V^{**} \cong V$ organized double-dual functionals in the linear functional setting above.

8 Eigenvalues and Eigenvectors

- **Definition**: For a linear operator $T : V \to V$ (or its matrix A), a nonzero vector $v \in V$ is an **eigenvector** with **eigenvalue** $\lambda \in F$ if $Tv = \lambda v$. That is, T acts on v by simple scalar stretching, with no change of direction. If an eigenvalue is not unique to an eigenvector, it is considered **degenerate**.

- **Characteristic Polynomial**: The eigenvalues of A are precisely the roots of the **characteristic polynomial** $\det(A - \lambda I) = 0$. This connects eigenvalues directly to the determinant: expanding $\det(A - \lambda I)$ gives a degree-n polynomial in λ whose roots are the eigenvalues, counted with **algebraic multiplicity**.

- **Eigenspace**: For a fixed eigenvalue λ, the set of all vectors satisfying $Av = \lambda v$ (equivalently $\ker(A - \lambda I)$) is a subspace called the **eigenspace** E_λ. Its dimension is the **geometric multiplicity** of λ, and it always satisfies $1 \leq \dim(E_\lambda) \leq$ (algebraic multiplicity of λ).

- **Diagonalization**: T is **diagonalizable** if V has a basis consisting entirely of eigenvectors of T. Relative to such a basis, $[T]$ is a diagonal matrix with the eigenvalues on the diagonal. This is the simplest possible matrix representative of T, in the same sense that the change-of-basis formula $[T]_{\mathfrak{B}'} = P^{-1}[T]_{\mathfrak{B}}P$ lets you seek the "best" basis for a given operator.

- **Trace and Determinant Identities**: $\text{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$, where the sums/products run over all eigenvalues counted with algebraic multiplicity. These give quick consistency checks and tie eigenvalues directly back to the two basis-independent (similarity-invariant) scalars you already know.

8.1 Simultaneous Eigenstates

- **Definition**: If two operators T and S on V share a common eigenvector v — that is, $Tv = \lambda v$ and $Sv = \mu v$ for some (possibly different) eigenvalues λ, μ — then v is called a **simultaneous eigenstate** (or simultaneous eigenvector) of T and S.

- **Commuting Operators**: A basis of simultaneous eigenvectors for T and S exists if and only if T and S **commute**, i.e. $TS = ST$. This is the key linear-algebra criterion: commutativity is exactly what allows two operators to be **simultaneously diagonalized** by the same basis. If v is a simultaneous eigenvector with $Tv = \lambda v$, $Sv = \mu v$, then $TSv = T(\mu v) = \mu\lambda v$ and $STv = S(\lambda v) = \lambda\mu v$, so $TSv = STv$ on every vector in a shared eigenbasis — and since a basis determines the operator, $TS = ST$ on all of V.

- **Tensor product connection**: For operators acting on different factors of a composite system, $T \otimes I$ and $I \otimes S$ (acting on $V \otimes W$) always commute automatically, since they act on independent "slots", so eigenstates of the form $v \otimes w$ (with $Tv = \lambda v$, $Sw = \mu w$) are automatically simultaneous eigenstates of $T \otimes I$ and $I \otimes S$. This is the standard way multi-particle systems acquire well-defined joint quantum numbers.

9 Inner Product and Hilbert Spaces

- At the beginning of our journey into linear algebra, we said that vectors need not have direction and magnitude to be considered a vector. We can add these properties by endowing one additional condition to our vector space: an inner product.

- You may have never heard of this term before, but you have certainly used it many times because our favorite vector calculus operator, the dot product, is an inner product.

- This is only one example, though, so let us explore further.

- **Inner Product:** Let F be the field of real numbers or the field of complex numbers, and V a vector space over F. An inner product on V is a function that assigns to each pair of vectors $\alpha, \beta \in V$, a scalar given by the operator function $< \cdot | \cdot >: V \times V \to F$ in such a way that for all $\alpha, \beta, \gamma \in V, c \in F$

  1. $< \alpha + \beta | \gamma > = < \alpha | \gamma > + < \beta | \gamma >$
  2. $\langle c\alpha | \beta \rangle = \langle \alpha | c\beta \rangle = c \langle \alpha | \beta \rangle$
  3. $\langle \beta | \alpha \rangle = \langle \beta | \alpha \rangle^*$, where the $(*)$ indicates complex conjugation.
  4. $\langle \alpha | \alpha \rangle > 0$, if $\alpha \neq 0$

- **Standard Inner Product (Dot Product)**: On F^n, and vectors $\alpha = (x_1, ..., x_n), \beta = (y_1, ..., y_n)$ by $\langle \alpha | \beta \rangle = \sum x_i \bar{y_i}$, which simplifies to be the standard dot product when $F = \mathbb{R}$.

- Another example of an inner product is the **Frobenius Inner Product**. For the vector space of real $n \times n$ matrices, we define $< A | B > = tr(A^T B)$.

- **Norm**: We define the magnitude or *norm* of a vector α to be $||\alpha|| = \sqrt{< \alpha | \alpha >}$.

- **Quadratic Forms:** is a homogeneous polynomial of degree two (multiplying all inputs by n, multiplies the polynomial by n^2). In simpler terms, it is a mathematical expression where every term is exactly the second power of a variable (e.g., X^2) or the product of two different variables (e.g., $x_1 x_2$). In the context of linear algebra, it is the function that assigns to each vector α the scalar $||\alpha||^2$.

- **Inner Product Space**: A vector space paired with a specified inner product. We call a finite-dimensional real inner product space a **Euclidean Space**, and the complex variant a **Unitary Space**

- **Orthogonality**: We say two vectors are orthogonal if their inner product is zero. If S is a set of vectors in V, S is called an **orthogonal set** provided that each vector is mutually orthogonal to each other. **Orthonormal Sets** are orthogonal sets with the condition that each vector in the set has norm of one.

- Orthonormal sets form great bases, making math very simple, and as such are the prime choice for studying quantum mechanics (we will be talking about these quite often). We can show any orthogonal set is linearly independent; we just need to ensure it spans our vector space.

- Starting with a basis in an inner product space, we can create from it an orthonormal basis using the **Gram-Schmidt Orthogonalization Process**:

  1. Given a linearly independent set $\beta = \{v_1, v_2, \ldots, v_n\}$, construct an orthogonal set $\{u_1, u_2, \ldots, u_n\}$
  2. Defining $u_1 = v_1$,
  3. For $k = 2, \ldots, n$,$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j$.
  4. If an orthonormal basis is desired, set $e_k = \frac{u_k}{\|u_k\|},\ k = 1, \ldots, n$.

- Let V be an inner product space and let S be any set of vectors in V. The orthogonal complement of S is the set $S^{\perp}$ of all vectors in V that are orthogonal to every vector in S.

- **Riesz Representation Theorem**: Let V be a finite-dimensional inner product space, and f a linear functional on V. Then there exists a unique vector β in V such that $f(\alpha) = \langle \alpha | \beta \rangle$ for all $\alpha \in V$. This theorem is the entire foundation of Dirac's Bra-ket notation, which will be discussed in the next lesson.

- For any linear operator T on a finite-dimensional inner product space V, there exists a unique linear operator $T*$ on V such that $\langle T\alpha | \beta \rangle = \langle \alpha | T^* \beta \rangle$. We call this operator the adjoint of T on V. (Note: the adjoint of an adjoint is the original matrix)

- Let V be a finite-dimensional inner product space and let $\beta = a_1, ..., a_n$ be an (ordered) orthonormal basis for V. Let T be a linear operator on V, and let A be the matrix of T in the ordered basis β. Then, $A_{kj} =< T\alpha_j | \alpha_k >$.

- **Properties of Adjoints**:

  1. $(T + U)^* = T^* + U^*$
  2. $(cT)^* = \bar{c}T^*$
  3. $(TU)^* = U^*T^*$
  4. $(T^*)^* = T$

- Consider T to be an endomorphism on V. If $T^* = T$, then we say T is **Hermitian**. If $T^* = T^{-1}$, then we say T is **Unitary**.

- **Properties of Hermitian Operators**

  1. All eigenvalues of a Hermitian Transformation are real
  2. Eigenvectors for distinct eigenvalues are orthogonal
  3. The determinant will be real if $F = \mathbb{C}$
  4. Diagonalize with unitary matrices with real diagonal

- **Properties of Unitary Operators**

  1. Preserves norms: $||Tx|| = ||x||$
  2. Preserves inner products: $\langle Tx | Ty \rangle =< x | y >$

3. All eigenvalues have modulus 1 ($\lambda = e^{i\theta}$)
4. Determinant will always have modulus 1

- **Spectral Theorem**: If A is Hermitian on V, then there exists an orthonormal basis V consisting of eigenvectors of A. Each eigenvalue of A is real.

- **Hilbert Spaces**: A (complex) vector equipped with an inner product and Cauchy Completeness, which mandates that each sum (even infinite) of vectors converges to an element within the space.

- One might ask why we need the Cauchy Completeness requirement. For finite dimensions, this condition is redundant, but for an infinite-dimensional Hilbert space, we need it. The best way to see why is to consider the vector space $\mathbb{R}[x]$, the set of all real polynomials. We can all agree that a sine wave isn't a polynomial, but if we were to take its Taylor expansion (an infinite series of polynomials that converge to a sine wave), we would see that it must be (but isn't) an element of the vector space. The Cauchy Completeness requirement patches this gap. This allows us to consider continuous bases in QM (like position and momentum).

10 Appendix: (To be completed)

10.1 Quotient Spaces

- We have discussed two ways to combine vector spaces: the tensor product and the direct sum. However, what if we want to "collapse" a vector space? To do so, we introduce the notion of Quotient spaces. But to discuss these, we first need to discuss what an equivalence relation is.

- **Relation**: For a set X, let $X \times X$ denote the set of all ordered pairs. A (binary) relation on X is a function on $R : X \times X \to \{0, 1\}$. In other words, R assigns to each ordered pair a one or zero; one if there is some desired relationship, and zero if there is not. Notation-wise, we often use $\sim$ to denote a relation between two numbers

- An example of a relation is the less-than relation. Take the set of all integers to be X here. In this case, the function R maps all pairs (a, b) where $a < b$ to one and every other pair to zero. Also, because this is such a widely used relation, we use the $<$ symbol instead of $\sim$

- **Equivalence Relation**: This is a specific type of relation that has similar properties to our algebraic equality. An *equivalence relation* is a relation that satisfies the following:

  1. Reflexive: $x \sim x, \ \forall \ x$ in X
  2. Symmetric: $y \sim x \iff x \sim y$
  3. Transitive: $x \sim y, y \sim z \implies x \sim z$

- You can quite easily see that equality falls under the umbrella of an equivalence relation, and less than does not. Another "more abstract example" is the equivalence relation of similar triangles in the x-y plane.

- We also saw another example of this when discussing modular arithmetic. The relation R_n is called **congruence modulo n** and occurs when you take the remainders of two numbers after dividing by n, and they are equal (e.g., for n=2, we have congruence of all odd numbers and congruence of all even numbers).

- **Equivalence Class (Coset)**: Because of the properties of equivalence relations, we can define sets of elements that all satisfy the equivalence relation in terms of one another and are maximal (no equivalent element left out). What we see is that all distinct *equivalence classes* are disjoint from one another (required else transitivity forces equality).

- Returning to the notion of congruence modulo n, for n=2, we see that the even numbers form an equivalence class and the odd numbers form another. Obviously, they are distinct because no even number is odd.

- With these prerequisites set, we can now discuss quotient spaces.

- **Quotient Space**: Let V be a vector space and W a subspace of V. The quotient space consists of equivalence classes (cosets) of the following form: $v + W = v + w : w \in W$. In other words, the quotient space is the set of all equivalence classes of the equivalence relation: $v \sim u \iff v - u \in W$, Thus, vectors that only differ by an element of W represent the same point. We can show this is a vector space, but we won't for the sake of brevity.

- An example is $V = \mathbb{R}^3$ and $W = \mathbb{R}^2 = \{xy - plane\}$. We can imagine $\mathbb{R}^3$ as an infinite stack of infinitely large sheets of paper (each parallel to the xy plane and representing W). V/W simply says that instead of considering a three-dimensional coordinate, we only care about the sheet of paper which we are on (e.g., the z coordinate).

- **Canonical Projection Map**: There is a natural linear map: $\pi : V \to V/W,\ \pi(v) = v + W$. This is called the quotient map or the canonical projection and simply linearly maps an element in V to its equivalence class.

- Note: the kernel/null space of the projection is precisely the subspace that was collapsed (for V/W, its W). Additionally, if you can't tell by virtue of the word projection, this map is surjective: $Im(\pi) = V/W$. We can easily write the dimension formula based on this using the rank-nullity theorem: $dim(V/W) = dim(V) - dim(W)$. Quotienting removes the dimensions contributed by W.

- If $\{w_1, ..., w_k, v_1, ..., v_m\}$ is a basis of V where $\{w_1, ..., w_k\}$ forms a basis of W, then $\{v_1 + W, ..., v_m + W\}$ forms a basis of V/W.

- **Universal Mapping Property of Quotient Spaces**: If $T : V \to U$ is linear and $W \subset ker(T)$, then there exists a unique linear map $\tilde{T} : V/W \to U$ such that $T = \tilde{T} \circ \pi$. In other words, every linear map that annihilates W factors uniquely through the quotient.

- First Isomorphism Theorem: For any linear map $T : V \to U$: $V/ker(T) \cong Im(T)$. This is arguably the most important theorem from quotient spaces. Interestingly, we can recover the rank-nullity theorem from considering the dimensions of this space.

- If you prefer to think about this in terms of direct sums. If $V = W \oplus U$, then $V/W \cong U$. So, quotient spaces sort of act as subtraction of vector spaces (not division, funnily enough).

10.2 Sequences and Chain Complexes

- Thus far, every structure we have explored has concerned itself with a single or a pair of spaces connected by a map. Chain complexes let us string together an entire sequence of vector spaces and maps in a coherent way. We want to see how the image of one map land relative to the kernel of the next.

- The answer to that question turns out to be a vector space in its own right (the homology of the complex); it measures, in a precise sense, how much a sequence "fails to be exact." Chain complexes were developed in algebraic topology to count holes in geometric shapes, but the purely linear-algebraic core is accessible right now, with exactly the tools we have built.

10.2.1 Sequences and Exactness

- **Chain Complex:** A sequence of vector spaces and linear transformations upon the vector spaces indexed by the integers, together with the fundamental requirement that consecutive maps compose to zero $d_{n-1} \circ d_n = 0, \forall\ n$:

$$\cdots \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$$

- We call the linear maps d_n boundary operators (or differentials). The condition that $d_{n-1} \circ d_n = 0,\ \forall n$ is incredible significant. This immediately tells us that the image of d_n is always contained in the kernel of d_{n-1}: $Im(d_n) \subset Ker(d_{n-1})$. The homology of the complex effectively measures how far each of these containments is from equality.

- We say that the chain complex is **exact** at C_n if $im(d_{n+1}) = ker(d_n)$. The complex is called an exact sequence if it is exact at each vector space. Exactness means that the sequence has no "slack". Whatever the incoming map hits is precisely what the outgoing map kills. Thus, exactness means no homology. You will see why in a moment.
- Exactness allows one space to be described entirely in terms of the image and kernel of neighboring maps. Failure of exactness measures the presence of new information, which is precisely what homology detects.
- There are two useful special cases

$$0 \to A \xrightarrow{f} B, \qquad A \xrightarrow{g} B \to 0$$

For the first case, exactness at A implies f is injective. For the second, exactness at B implies g is surjective.

10.2.2 Short Exact Sequences

- **Short Exact Sequences**: Exact sequences of the form: $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$
- A few of the most important properties are that automatically f is injective, g is surjective, ker(g)=im(f). This means f embeds A into B and g presents C as the quotient of $B/im(f) \cong C$
- Splitting Lemma: Over a field F, every short exact sequence splits: $B \cong A \oplus C$. This is a special feature of vector spaces that doesn't hold for general modules over a ring (a generalized form of vector fields, but for rings not fields). The splitting says that B decomposes into a copy of A (sitting inside via f) and a complementary copy of C (projecting out via g). The rank-nullity theorem is essentially a restatement of this splitting for a single linear map.
- **The Fundamental Quotient Short Exact Sequences**: If W is a subspace of V, then there is a canonical short exact sequence:

$$0 \to W \xrightarrow{i} V \xrightarrow{\pi} V/W \to 0$$

- **First Isomorphism Theorem** as a Short Exact Sequence: Using properties of SES, we can prove the First Isomorphism Theorem for an arbitrary linear map $T : V \to U$. There exists a natural SES:

$$0 \to ker(T) \xrightarrow{i} V \to im(T) \to 0$$

From above, we know that $V/im(i) = V/ker(T) \cong im(T)$.

10.2.3 n-Chains, Cycles, and Boundaries

- Fix an index n. We give names to the three key subspaces of C_n:
  - **n-Chains**: Every element of C_n is an n-chain; this is just a formal way to say the elements at position n in the sequence.
  - **n-Cycles**: $Z_n = ker(d_n) \subset C_n$. These are the chains that the outgoing boundary operator kills. Geometrically, they have "no boundary," meaning that these are like closed loops (e.g. tori have no clear boundary).
  - **n-Boundaries**: $B_n = im(d_{n+1}) \subset C_n$. These are the chains that arise as the boundary of an (n+1)-chain.
- Something to note: In a chain complex, the indices go from higher to lower because the boundary operator lowers the dimension.
- By properties of the compositions of functions in a chain complex, we have $B_n = im(d_{n+1}) \subset ker(d_n) = Z_n$. Thus, every boundary is automatically a cycle, but not every cycle needs to be a boundary. This is the notion of slack in the sequence we mentioned earlier.
- These three subspaces fit together as a chain: $B_n \subset Z_n \subset C_n$. Both Z_n and B_n are subspaces of C_n, and B_n is a subspace of Z_n, so the quotient Z_n/B_n

10.2.4 Homology

- **Homology** (at position n): The quotient vector space: $H_n = Z_n/B_n = ker(d_n)/im(d_{n+1})$. Its dimension is called the **n-th Betti number** of the complex, β_n. Two cycles represent the same homology class iff their difference is a boundary $z - z' \in B_n \implies [z] = [z']$, or we say z and z' are homologous. If every cycle is a boundary at position n, the complex is exact ($H_n = 0$).
- In summary $H_1 = Z_1/B_1$, where Z_1 are all closed loops (cycles), B_1 are all the loop that are boundaries of filled-in surfaces, and H_1 is the remainder after declaring all boundary loops to be zero.
- Homology does not count loops; it counts independent ways of having loops that cannot be filled in.

10.2.5 The Euler Characteristic

- For a finite chain complex (nonzero in only finitely many degrees), define the Euler characteristic: $\chi = \sum_n (-1)^n dim(C_n)$. The remarkable fact (from the rank nulli-y theorem applied at each stage) is that χ can also be computed from the homology:

$$\chi = \sum_n (-1)^n dim(H_n) = \sum_n (-1)^n \beta_n$$

- This equality holds regardless of the specific maps, depending only on the dimensions involved. It is an early example of the principle that topological invariants (the left side, counting dimensions of the chain spaces) agree with algebraic invariants (the right side, counting dimensions of the homology groups).

10.2.6 Chain Maps

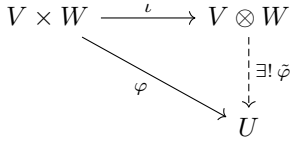
- **Chain Map**: Given two chain complexes $(C_\bullet, d^C), (D_\bullet, d^D)$, a chain map $f_\bullet : C_\bullet \to D_\bullet$ is a collection of linear maps $f_n : C_n \to D_n$ such that the following commutes for every n: $d_n^D \circ f_n = f_{n-1} \circ d_n^C$.
- This simply means that it doesn't matter whether you apply the boundary operator before or after applying f_n; you end at the same place. Chain maps are the morphisms (shout out to category theory) of chain complexes; they are the structure-preserving maps in this category. (In the same way, linear transformations are the morphisms between vector spaces)
- A chain map $f_\bullet$ induces a well-defined linear map on homologies: $f_* : H_n(C) \to H_n(D), \quad [z] \mapsto [f_n(z)]$. This is the payoff of the $d_{n-1} \circ d_n = 0$ condition./. It is precisely what allows maps between complexes to descend to maps between homology spaces (allowing us to compare more "fundamental" spaces).

10.3 Universal Mapping Properties

- By this point, we have defined a lot of spaces: the tensor product, the direct sum, the dual space, the quotient spaces, etc. Each time, the definition came with a list of rules and a basis for computation. But there is a cleaner, more unified way to say what all of these spaces are, by specifying what maps out of (or into) them look like. This is the idea of a universal mapping property.
- **Universal Mapping Property (UMP)**: A construction satisfies a *universal mapping property* if it comes equipped with a canonical map (into or out of it) such that every other map of the same type "factors" (can be expressed as the composition of maps) through it uniquely.
- The word unique is doing all the work here. The UMP doesn't just say that factoring exists; it says there is exactly one way to factor, which pins down the space up to isomorphism. This shows the power of the UMP as a tool; it is general enough to be used as a general tool, but small enough that it isn't too "basic" that you can't do anything with it. Additionally, when proving a UMP, you have to ensure that you can prove both existence and uniqueness (sometimes uniqueness is easier).

10.3.1 UMP of Tensor Products

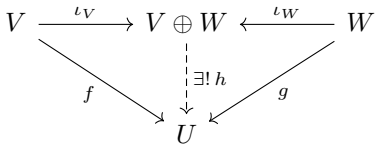
- **UMP of the Tensor Product:** $V \otimes W$ is the unique vector space (up to isomorphism) equipped with a bilinear map $\iota : V \times W \to V \otimes W,\ (v, w) \mapsto v \otimes w$, such that for every vector space U and every bilinear map $\varphi : V \times W \to U$, there exists a *unique* linear map $\tilde{\varphi} : V \otimes W \to U$ satisfying $\tilde{\varphi}(v \otimes w) = \varphi(v, w)$.



  - In other words, bilinear maps *out of* $V \times W$ are in exact correspondence with linear maps *out of* $V \otimes W$. The tensor product is the universal linearization of bilinear data.
  - This is precisely what we stated in Section 7, but now we can recognize it as an instance of a general pattern rather than an ad hoc definition.

10.3.2 UMP of Direct Sum

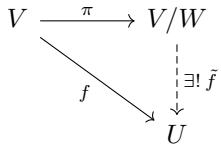
- **UMP of the Direct Sum:** $V \oplus W$ is characterized by the existence of inclusion maps $\iota_V : V \to V \oplus W$ and $\iota_W : W \to V \oplus W$ such that for every vector space U and linear maps $f : V \to U$, $g : W \to U$, there is a unique linear map $h : V \oplus W \to U$ with $h \circ \iota_V = f$ and $h \circ \iota_W = g$.



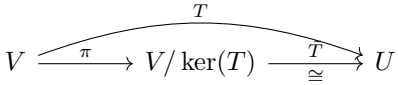
  - Dually, $V \oplus W$ also satisfies a UMP for maps *into* it via projections π_V, π_W, making it both a *coproduct* and a *product* in the category of vector spaces — a lucky coincidence of working over a field.
  - The direct sum is "just big enough" to receive maps from both V and W independently, without imposing any relation between them.

10.3.3 UMP of the Quotient Space

- **UMP of the Quotient Space:** Given a subspace $W \subseteq V$, the quotient V/W is characterized by the canonical projection $\pi : V \to V/W$ and the property that any linear map $f : V \to U$ with $W \subseteq \ker(f)$ factors uniquely through π: there is a unique $\tilde{f} : V/W \to U$ with $f = \tilde{f} \circ \pi$.



  - This says V/W is the "most general" space you can map V into while killing W. Any other such space receives a map from V/W.
  - This also clarifies why the first isomorphism theorem holds: a surjective linear map $T : V \to U$ with kernel W factors as $V \xrightarrow{\pi} V/W \xrightarrow{\tilde{T}} U$, where $\tilde{T}$ is now an isomorphism.



10.3.4 Uniqueness from the UMP:

- Any two spaces satisfying the same UMP are canonically isomorphic. Suppose P and P' both satisfy the UMP for $V \otimes W$. Then by P's UMP, the canonical bilinear map of P' factors uniquely through P, giving $f : P \to P'$. Symmetrically, P's canonical map factors uniquely through P', giving $g : P' \to P$. The composites $g \circ f$ and $f \circ g$ must each be the identity by the uniqueness clause (since the identity also works). So $P \cong P'$.

- This is the real power of the UMP: we don't need to check that two constructions produce the "same" space element-by-element. We just need to check they satisfy the same universal property, and isomorphism follows for free.

- The UMP is a shift in perspective, from "what are the elements of this space?" to "how does this space interact with all other spaces?" It turns out the latter question determines the space completely.

10.4 Canonical Forms

- In Section 8, we said that a linear operator $T : V \to V$ is diagonalizable if there exists a basis of eigenvectors giving $[T]_\beta$ a diagonal matrix with eigenvalues on the diagonal. This is the simplest possible matrix representative for T. But not every operator is diagonalizable. Canonical forms answer the question: what is the simplest matrix that we can always achieve, regardless of whether T diagonalizes?

- The key insight is that "simplest" depends on what we allow. If we work over an algebraically closed field (like $\mathbb{C}$) and allow any invertible change of basis, we get the **Jordan Canonical Form**. If we only care about the rational structure and want to stay inside the base field, we get the **Rational Canonical Form**. And if we are willing to use two separate bases (one for the domain, one for the codomain), we can always achieve the **Elementary Canonical Form** for any linear map between two vector spaces.

10.4.1 Elementary Canonical Form

- So far, canonical forms for operators have required eigenvalues and a square matrix. But, what is the simplest form for a general linear map $T : V \to W$ between different vector spaces, where we can choose any basis for V and W.

- We have already seen a way to reduce systems of equations using a type of canonical form, namely RREF. The criterion was that we did not change the solution space of the transformation. The Elementary Canonical Form broadens this categorization (in exchange for altering the solution space)

- **Elementary Canonical Form**: Let $T : V \to W$ be a linear map with $dim(V) = n$, $dim(W) = m$, and $rank(T) = r$. Then there exists ordered bases $\beta \subset V, \gamma \subset W$ such that the matrix $[T]^\gamma_\beta = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$

- Here, I_r is hte $r \times r$ identity and all other entries are zero. This is the ECF of T. The construction is straightforward: choose a basis $\{v_1, ..., v_r\}$ for the $ker(T)$ in V, and extend to a full basis of V using $ker(T)$. Then, we set $w_i = Tv_i,\ i \leq r$ and extend $\{w_1, ..., w_r\}$ to a full basis of W. In these bases, T sends the first r basis elements to the first r basis vectors and then kills the rest.

- This is essentially the rank-nullity theorem made visible in a matrix. The rank r counts the ones on the diagonal, the nullity $n - r$ counts the zero columns, and the remaining $m - r$ zero rows count the "unused" dimensions of W.

- The ECF tells us that, up to independent changes of basis on the domain and codomain, every linear map looks the same up to its rank. There is no finer invariant available.

- In particular, two matrices A, B are equivalent in this canonical form if they are related by $B = QAP^{-1}$ for invertible P, Q iff they have the same rank. This is equivalent to saying the ECF holds that matrices are equivalent if you can put one matrix into another using both elementary row and column operations.

10.4.2 Jordan Canonical Form

- When we restrict to operators $T : V \to V$ and ask for a basis in which $[T]$ is as simple as possible, the obstruction to diagonalization turns out to be repeated eigenvalues whose eigenspaces are "too small." The Jordan Canonical Form handles this issue precisely.

- Generalized Eigenvectors: For an eigenvalue λ of T, the generalized eigenspace is $K_\lambda = ker((T - \lambda I)^n)$ for $n = dim(V)$. Every vector K_λ eventually maps to zero under repeated application of $(T - \lambda I)$, even if it's not itself an eigenvector.

- The ordinary eigenspace $E_\lambda = ker(T - \lambda I) \subset K_\lambda$. If the dimensions are equal, then λ contributes no obstruction and can be handled by diagonalization alone. If the dimension of E_λ is less than that of K_λ, the "missing" eigenvectors are exactly what force the off-diagonal 1s in the Jordan blocks. For more explanation, feel free to look up a full proof.

- **Jordan Block:** A $k \times k$ Jordan block for eigenvalue λ is the matrix below, with λ on the diagonals and 1s on the superdiagonal:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix}$$

- A cool thing to see: A 1×1 Jordan block is just λ, so diagonal matrices are a special case where every block has size one.

- **Jordan Canonical Form**: Let V be a finite-dimensional vector space over an algebraically closed field (in the real field, we may obtain imaginary eigenvalues) F, and let $T : V \to V$ be an operator. Then, there exists a basis β for V such that we have a block diagonal matrix (representing our transformation) of Jordan blocks:

$$[T]_\beta = \begin{pmatrix} J_{k_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{k_r}(\lambda_r) \end{pmatrix}$$

- This is the JCF of T, and it is unique up to reordering of the blocks. The eigenvalues are simply the ones obtained by the characteristic polynomial. The field, as mentioned, must be algebraically closed such that the eigenvalues $\lambda \in F$. Note: the number of Jordan blocks for a given λ is $dim(E_\lambda) = dim(ker(T - \lambda I)$, the geometric multiplicity. The sum of block sizes for λ equals $dim(K_\lambda)$, the algebraic multiplicity. These two numbers only agree if T is diagonalizable at λ

- Reading off the block structure: The sizes of the Jordan blocks for λ can be recovered from the sequence of ranks: $rank(T - \lambda I)^0,\ rank(T - \lambda I)^1,\ rank(T - \lambda I)^2, ...$

- Why does the JCF matter? It gives the most refined picture of how T acts on V. Computing powers T^n, matrix exponentials e^T, or the long-run behavior of a discrete dynamical system $x_{k+1} = Tx_k$ is straightforward in Jordan form and often difficult in a generic basis. Each Jordan block $J_k(\lambda)$ contributes terms of the form $\lambda^n \binom{n}{j}$ when computing matrix powers and $e^{\lambda t} \frac{t^j}{j!}$ when computing matrix exponentials. The off-diagonal 1's in the Jordan blocks are precisely what generate these polynomial prefactors, encoding the effects of repeated eigenvalues and generalized eigenvectors.

10.4.3 Rational Canonical Form

- The JCF requires the characteristic polynomial to split completely over F. If we work over $\mathbb{R}$ and T has complex eigenvalues, the JCF doesn't exist as a real matrix. The Rational Canonical form resolves this within the base field, at the cost of less diagonal structure.

- **Invariant Factors:** The Rational Canonical Form (RCF) is built from the *invariant factors* of T: a sequence of monic polynomials $f_1 \mid f_2 \mid \cdots \mid f_k$, where each polynomial divides the next and $f_1 f_2 \cdots f_k = \chi_T(x)$, the characteristic polynomial of T. The largest invariant factor, f_k, is the *minimal polynomial* of T, the unique monic polynomial of least degree such that $f_k(T) = 0$.

- **Companion Matrix:** For a monic polynomial $f(x) = x^k + c_{k-1}x^{k-1} + \cdots + c_1 x + c_0$, the companion matrix of f is

$$C(f) = \begin{pmatrix} 0 & & & & -c_0 \\ 1 & 0 & & & -c_1 \\ & 1 & \ddots & & -c_2 \\ & & \ddots & \ddots & \vdots \\ & & & 1 & -c_{k-1} \end{pmatrix},$$

  whose characteristic polynomial is exactly f. It is the canonical way to realize a polynomial as a matrix.

- **Rational Canonical Form (RCF):** There exists a basis for V such that $[T]$ is block-diagonal with companion matrices $C(f_1),\ C(f_2),\ \ldots,\ C(f_k)$ on the diagonal. This form exists over *any* field F and requires no assumption that F be algebraically closed.

- The RCF and JCF contain exactly the same information and are related by factoring the invariant factors into linear factors over the algebraic closure $\overline{F}$. Over $\mathbb{C}$, the two forms are equivalent. Over $\mathbb{R}$, the RCF packages complex-conjugate pairs of Jordan blocks into real 2×2 companion blocks, keeping the representation entirely over $\mathbb{R}$.

- **The Cayley–Hamilton theorem**: that every matrix satisfies its own characteristic polynomial—is an immediate consequence of the RCF. The characteristic polynomial is the product of the invariant factors, and each companion block $C(f_i)$ satisfies $f_i(C(f_i)) = 0$. Consequently, the product $\chi_T(T) = f_1(T) \cdots f_k(T)$ annihilates each invariant subspace associated with the companion blocks and therefore annihilates all of V: $\chi_T(T) = 0$.

10.4.4 Singular Value Decomposition

- While diagonalization requires a square matrix with a full eigenbasis, the **Singular Value Decomposition** generalizes this to any $m \times n$ matrix, regardless of shape or rank.

- **Singular Value Decomposition** For any $m \times n$ matix A over $\mathbb{R}$ or $\mathbb{C}$, there exists matrices U, Σ, V such that: $A = U\Sigma V^\dagger$.

- Here, U is an $m \times m$ unitary matrix (we call the columns of this matrix **left singular vectors**). Σ is an $m \times n$ matrix with non-negative real entries ($\sigma_1 \geq \sigma_2 ... \geq \sigma_r > 0$) along the diagonals (with zeros everywhere else). V is an $n \times n$ unitary matrix (columns are **right singular vectors**). Moreover, we call the diagonal elements of A the singular values, and r = rank (A).

- Here is the general structure of the proof: Consider $A^T A$. As a positive semidefinite Hermitian matrix, the Spectral Theorem guarantees an orthonormal eigenbasis $\{v_1, ..., v_n\}$ with eigenvalues $\lambda_1 \geq ... \lambda_n \geq 0$. Set $\sigma_i = \sqrt{\lambda_i}$ and define $u_i = (1/\sigma_i) A v_i$ for $i \leq r$; one can then verify that the set of u_i's are orthonormal. We can then extend by Gram-Schmidt to a full orthonormal basis of $\mathbb{R}^m$, collect into U, and verify $A = U\Sigma V^\dagger$.

- Let's now make sense of what this means: Every linear transform A decomposes into three steps: $V^\dagger$ rotates and reflects the input so the right singular vectors align with the standard basis, σ scales each axis by the singular values (directions outside the row space are zeroed), and U rotates/reflects the result into the output singular vector basis of the u_i's.

- This is why every matrix is, up to an orthonormal change of basis, just a coordinate-wise scaling.
- From our proof, we see that σ_i^2 relates singular values to eigenvalues. For a normal matrix: $\sigma_i = |\lambda_i(A)|$. For a positive semidefinite Hermitian matrix, the SVD and the spectral decomposition are the same.
- Unlike diagonalization, the SVD exists for every matrix
- The SVD provides explicit orthonormal bases for all subspaces of A.
  - Column space: spanned by $u_1, ..., u_r$ (first r columns of U)
  - **Left null space** (the set of y such that $y^T A = 0$): spanned by $u_{r+1}, ..., u_m$ (remaining columns of U)
  - Row space: spanned by $v_1, ..., v_r$ (first r columns of V)
  - Null space: spanned by $v_{r+1}, ..., v_n$ (remaining columns of V)
- **The Moore-Penrose Pseudoinverse**: For an SVD, $A = U\Sigma V^\dagger$, define the pseudoinverse $A^+ = V\Sigma^+ U^\dagger$, where Σ^+ replaces each nonzero σ_i with $1/\sigma_i$ and takes the transpose. This generalizes the matrix inverse to a non-square or rank-deficient matrix. $x = A^+ b$ gives the minimum-norm least squares solution to $Ax = b$ and is the unique solution that also lies in the row space of A.

10.5 Multilinear and Alternating Forms

- We already briefly touched on these forms earlier when discussing the tensor and the determinant, as their derivations are reliant on these fundamental objects.
- **Multilinear Form**: A multilinear form (or k-form) on a vector space V is a map $\omega : V^k \to F$ that is linear in each argument separately, holding the others fixed. When $k = 1$, this is simply a linear functional; when k=2, it is a bilinear form (as used in section 7).
- **Alternating Form:** A k-form ω is alternating iff it vanishes whenever two of its arguments are equal $\omega(..., v, ..., v, ...) = 0$. This forces ω to change sign under any swapping of two arguments: $\omega(..., v_i, ..., v_j, ...) = -\omega(..., v_j, ..., v_i, ...)$. Antisymmetry is a consequence of the alternating condition (for $char(F) \neq 2$). A set of k vectors is linearly dependent iff every alternating form vanishes on them. (This is why determinants of square matrices can be used to detect linear independence).
- The space of all alternating k-forms on an n-dimensional space V is denoted $\Lambda^k(V^*)$, the **k-th exterior power of** V^*. Its dimension is $\binom{n}{k}$ (since it asks how many ways you can choose and organize k vectors (in a vector space of dimension n) such that they are all linearly independent (you are choosing k basis vectors).
- **The Wedge Product**: Given forms $\omega \in \Lambda^j(V^*)$ and $\eta \in \Lambda^k(V^*)$, their wedge product (or exterior product) $\omega \wedge \eta \in \Lambda^{j+k}(V^*)$ is defined by antisymmetrizing the tensor product:

$$(\omega \wedge \eta)(v_1, \dots, v_{j+k}) = \frac{1}{j!\,k!} \sum_{\sigma \in S_{j+k}} \operatorname{sgn}(\sigma)\, \omega\big(v_{\sigma(1)}, \dots, v_{\sigma(j)}\big)\eta\big(v_{\sigma(j+1)}, \dots, v_{\sigma(j+k)}\big).$$

- The wedge product is associative and bilinear, and satisfies $\omega \wedge \eta = (-1)^{jk} \eta \wedge \omega$. Particularly, $\omega \wedge \omega = 0$ for any odd-degree form.
- If $\{f_1, ..., f_n\}$ is the dual basis of V^*, then the wedge products $\{f_{i_1} \wedge \cdots \wedge f_{i_k} : i_1 < \cdots < i_k\}$ form a basis for $\Lambda^k(V^*)$
- We already mentioned that the determinant is a direct construction from alternating (multilinear) forms, but we should reiterate it to show the importance of these forms. Additionally, k-forms re-emerge in differential geometry in the study of differential forms.
- **Pullback**: If $T : V \to W$ is linear and $\omega \in \Lambda^k(W^*), the pullback\ T^*\omega \in \Lambda^k(V^*)$ is defined by $(T^*\omega)(v_1, ..., v_k) = \omega(Tv_1, ..., Tv_k)$. For $k = n = dim(V) = dim(W)$, this gives $T^*(\det_W) = \det(T) \cdot \det_V$, recovering the change-of-variables formula as a special case.

11 Conclusion

While not the most in-depth and rigorous dive into Linear Algebra, I hoped this served as a good primer for the material within the subject. As an undergraduate, I often wished I had an intuitive understanding of concepts in my Algebra classes before having them rigorously defined. Hopefully, this served as the bridge to cross the gap I once had. Please let me know if there are any mistakes or if there is any useful material you wish to be covered. Thank you so much for reading!